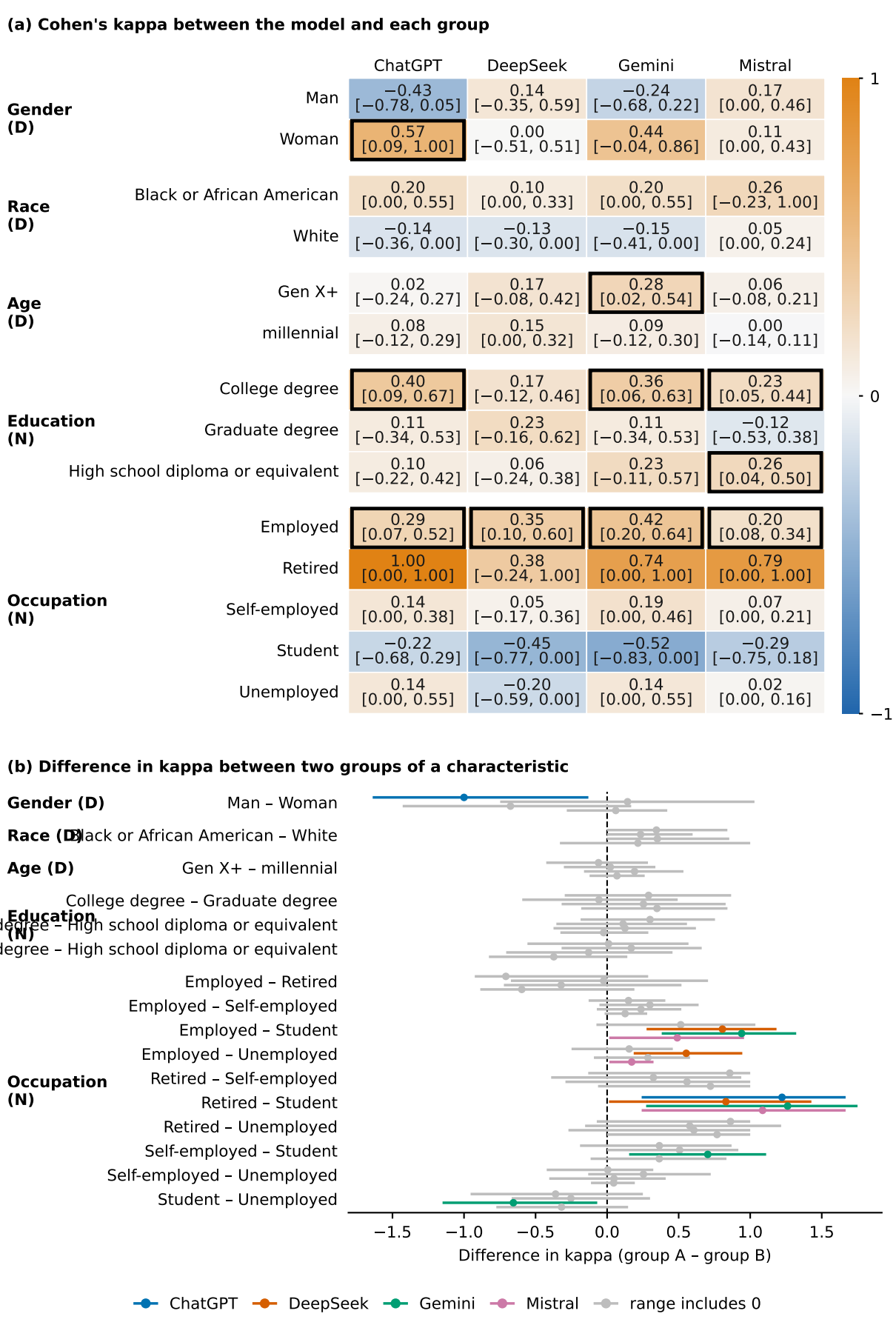
(a) Cohen's kappa between the model and each group
ChatGPT
DeepSeek
Gemini
Mistral
Gender (D)
Man
−0.43 [−0.78, 0.05]
0.14 [−0.35, 0.59]
−0.24 [−0.68, 0.22]
0.17 [0.00, 0.46]
Woman
0.57 [0.09, 1.00]
0.00 [−0.51, 0.51]
0.44 [−0.04, 0.86]
0.11 [0.00, 0.43]
Race (D)
Black or African American
0.20 [0.00, 0.55]
0.10 [0.00, 0.33]
0.20 [0.00, 0.55]
0.26 [−0.23, 1.00]
White
−0.14 [−0.36, 0.00]
−0.13 [−0.30, 0.00]
−0.15 [−0.41, 0.00]
0.05 [0.00, 0.24]
Age (D)
Gen X+
0.02 [−0.24, 0.27]
0.17 [−0.08, 0.42]
0.28 [0.02, 0.54]
0.06 [−0.08, 0.21]
millennial
0.08 [−0.12, 0.29]
0.15 [0.00, 0.32]
0.09 [−0.12, 0.30]
0.00 [−0.14, 0.11]
Education (N)
College degree
0.40 [0.09, 0.67]
0.17 [−0.12, 0.46]
0.36 [0.06, 0.63]
0.23 [0.05, 0.44]
Graduate degree
0.11 [−0.34, 0.53]
0.23 [−0.16, 0.62]
0.11 [−0.34, 0.53]
−0.12 [−0.53, 0.38]
High school diploma or equivalent
0.10 [−0.22, 0.42]
0.06 [−0.24, 0.38]
0.23 [−0.11, 0.57]
0.26 [0.04, 0.50]
Occupation (N)
Employed
0.29 [0.07, 0.52]
0.35 [0.10, 0.60]
0.42 [0.20, 0.64]
0.20 [0.08, 0.34]
Retired
1.00 [0.00, 1.00]
0.38 [−0.24, 1.00]
0.74 [0.00, 1.00]
0.79 [0.00, 1.00]
Self-employed
0.14 [0.00, 0.38]
0.05 [−0.17, 0.36]
0.19 [0.00, 0.46]
0.07 [0.00, 0.21]
Student
−0.22 [−0.68, 0.29]
−0.45 [−0.77, 0.00]
−0.52 [−0.83, 0.00]
−0.29 [−0.75, 0.18]
Unemployed
0.14 [0.00, 0.55]
−0.20 [−0.59, 0.00]
0.14 [0.00, 0.55]
0.02 [0.00, 0.16]
1
0
−1
(b) Difference in kappa between two groups of a characteristic
Gender (D)
Man – Woman
Race (D)
Black or African American – White
Age (D)
Gen X+ – millennial
Education (N)
College degree – Graduate degree
degree – High school diploma or equivalent
degree – High school diploma or equivalent
Occupation (N)
Employed – Retired
Employed – Self-employed
Employed – Student
Employed – Unemployed
Retired – Self-employed
Retired – Student
Retired – Unemployed
Self-employed – Student
Self-employed – Unemployed
Student – Unemployed
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Difference in kappa (group A – group B)
ChatGPT
DeepSeek
Gemini
Mistral
range includes 0